World Drift or Question Drift?

Question Identity as a Precondition for Continual World Model Evaluation

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Abstract

Continual world model evaluation usually compares predictions across checkpoints as though they answered a fixed question. In an end-to-end model, however, that question may be distributed across conditioning history, intervention, target, horizon, and grounding. These components can change without creating a new output slot. We call this *unversioned question drift*. Once the earlier specification is unavailable, a changed prediction alone cannot distinguish world change, question change, and estimation change. On a pre-specified 12-query stream, identity-blind learners retain live-only scores below 0.001 while bounded historical-query classification error reaches 0.68. A query-conditioned replay control retains near-zero historical error. We formalize versioned question specifications, scoped continuity probes, and an audit record for deciding whether outputs from different checkpoints are comparable.

1 Introduction: Unversioned Question Drift

Consider a deployed robot that uses an action-conditioned world model to forecast collisions along planned rollouts. At two checkpoints, a collision probability recorded under the same label changes from 0.1 to 0.8. This difference may reflect changed corridor dynamics, estimation change, or a silent switch from a 10-step left-turn rollout to a 30-step right-turn rollout. Without the conditioning question, the two values cannot yet be interpreted as revisions of the same knowledge.

Classical general value function (GVF) formulations make policy, cumulant, and continuation explicit [9, 19, 24]. They therefore provide a clean predictive-question case when those components and their referenced implementations are immutable or versioned. End-to-end world models can distribute the same information across shared heads, generated conditioning, rollout horizons, and sensor or token interpretations. Joint training can collapse conditioning, learned mechanisms can generate predictive questions themselves [23], and distillation or merging can discard output provenance.

Position. A continual world model should retain question specifications alongside parameters and predictions. The specification determines whether two outputs remain comparable over time. Without it, low current error can coexist with failure on earlier questions, and a numerical change can be mistaken for adaptation when the question itself has changed. We therefore treat question identity as an evaluation prerequisite. Before comparing checkpoints, an evaluator should resolve question identity and lineage, declare the comparison scope, and abstain when the specification or reference anchor is unavailable. This is not an architectural prescription. It makes question continuity a testable evaluation dimension.

Continual-learning metrics assume stable task-column semantics [4, 6, 14], and continual RL and world model methods study adaptation and retention under that assumption [11, 13, 28]. World or concept drift changes the data-generating relation while the predictive question stays fixed [5, 7]; question drift changes its intervention, target, or temporal semantics. Unversioned question drift is

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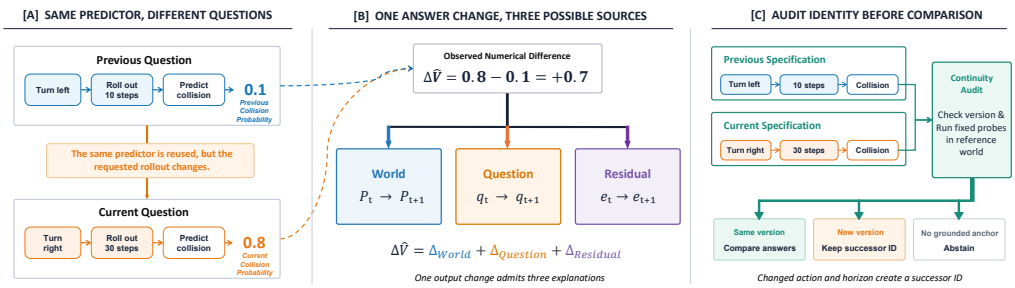


Figure 1: A checkpoint difference is comparable only after versioned specifications establish question identity. Probes test scoped compatibility; attribution comes later. The drift split is conceptual, not an identifiable causal decomposition.

the latter case without a new task slot or version. Task-conditioned architectures provide identity handles [15, 16, 18]; versioned concept memory and decision-centric evaluation address adjacent definition and use-alignment problems [12, 29]. Unlike task or context detection, it can occur with an unchanged observed behavior stream: a planner, decoder, or learned interface changes the rollout contract without a new task boundary. The prior question is: do two checkpoint outputs still answer the same predictive question?

2 Problem Formulation: Predictive Questions and Identity

Let $H_t \in \mathcal{H}$ be sensorimotor history and P the environment process. A predictive question is $q = (\kappa, u, Y, \tau, \iota)$, where $Y = (Z, F)$ specifies a target random object and a declared readout. Its predictive law and reported answer are

$$\mathfrak{L}(P, q; h) = \mathcal{L}_P^{u,\iota}(Z_q \mid \kappa(H_t) = \kappa(h)), \qquad \mathcal{A}(P, q; h) = F(\mathfrak{L}(P, q; h)). \tag{1}$$

We write $Z_q = Z(\tau)$ for the target random object specified by Z, with temporal scope τ. It can be a finite-horizon trajectory, continued return, event outcome, or another random object. The law $\mathcal{L}_P^{u,\iota}$ is induced by environment P, intervention kernel u, and interpretation map ι. Here κ selects history, u may be an action script or history-dependent policy, ι interprets tokens, channels, and actions, and F returns the law, an event probability, an expectation, or another statistic. We put the requested intervention u in q and actuator physics in P; this declared boundary remains fixed [25]. The externally versioned semantic contract is q, even if learned modules implement it. A change to declared meaning revises q; a numerical update that preserves it changes θ.

Several current systems expose components through which question continuity could be tracked, although this does not show that drift occurs in those systems. DreamerV3 exposes history, imagined actions, and prediction heads [8]. DIAMOND and Navigation World Models condition future video on histories and actions [1, 2]. Action-conditioned prediction supports representation learning [30], while WorldTrace changes which history remains addressable [27].

GVFs provide an analytically transparent instance of Eq. (1): the target law is a continued return and F returns its expectation. The Alberta Plan and classical GVF formulations motivate treating predictive questions as explicit objects. Here, ι extends that perspective to learned interfaces whose token, channel, or action semantics can change across checkpoints [17, 22].

Question identity. Two reports share identity only if they reference the same immutable specification and version. Any specification change creates a successor ID, even when the current answers coincide.

Probe compatibility. Probe compatibility asks a narrower question: whether two distinct specifications are locally interchangeable. Given a fixed reference environment P_{ref}, a finite probe set $B = \{h_i\}_{i=1}^m$, audit functionals ϕ_q into a shared metric space, a symmetric discrepancy D_s, and

tolerance $\epsilon \geq 0$, define

$$\widehat{d}_{B,\phi}^{P_{\mathrm{ref}}}(q, q') = \frac{1}{m} \sum_{i=1}^{m} D_s(\phi_q(\mathfrak{L}(P_{\mathrm{ref}}, q; h_i)), \phi_{q'}(\mathfrak{L}(P_{\mathrm{ref}}, q'; h_i))) \,. \tag{2}$$

This test is pairwise and scope-dependent. It is not an equivalence relation and never merges IDs. The audit functional ϕ need not equal the production readout F. Passing the test therefore licenses only the declared projections in P_{ref}, such as registered shared horizons [20]; it does not license arbitrary production answers or transfer to a new environment. Cross-environment claims require a pre-registered environment suite and a worst-case criterion. The displayed $\widehat{d}_B$ is a point estimate. A bootstrap interval $[L, U]$ over predeclared independent probe units determines the audit outcome: compatible if $U \leq \epsilon$, incompatible if $L > \epsilon$, and unresolved otherwise. The resampling unit and rollout count must be recorded. If the declared reference anchor—a retained environment, simulator, or suite—is unavailable, the protocol abstains whenever compatibility must be established.

3 Theoretical Analysis: Ambiguous Sources of Prediction Change

For a common anchor history h, let $V(P, q; h)$ be a scalar functional of $\mathcal{A}(P, q; h)$. Let $\hat{V}_t = \hat{V}_{\theta_t}(q_t; h)$ be the estimator under the declared question contract, and let $e_t = \hat{V}_t - V(P_t, q_t; h)$. The observed change $\Delta\hat{V}_t := \hat{V}_{t+1} - \hat{V}_t$ admits the following telescoping decomposition:

$$\Delta\hat{V}_t = \underbrace{V(P_{t+1}, q_t; h) - V(P_t, q_t; h)}_{\text{world drift}} + \underbrace{V(P_{t+1}, q_{t+1}; h) - V(P_{t+1}, q_t; h)}_{\text{question drift}} + \underbrace{e_{t+1} - e_t}_{\text{residual change}} \,. \tag{3}$$

Equation (3) is diagnostic, not an estimator. The split depends on the telescoping order; here the question term is evaluated in P_{t+1}. The residual is not a pure parameter effect because changing P or q can change the error of a fixed predictor. A learner update is isolated only when P, q, and h are held fixed. If the compared question is unknown, the numerical difference remains computable but cannot support claims about adaptation, forgetting, or knowledge revision.

Consider an identity-blind observer that receives only $O_t = (\hat{V}_{0:t}, H_{0:t}^{\mathrm{beh}}, A_{0:t}^{\mathrm{beh}})$, not the target-question specification or its earlier certificate.

Proposition 1 (No valid identity-blind comparison in general)**.** *If the earlier question certificate is unavailable, no decision rule based only on O_t can be universally correct over any model class containing the three constructions below. Such a rule cannot always decide whether two reports revise the same predictive knowledge or attribute their difference to world, question, or estimation change.*

Construction. Use one recurrent state, an observed behavior action a_0, and unobserved queried actions a_L, a_R. Hold the behavior outcome fixed and let the report change from 0 to 1. The same O_t arises in three cases: (A) P is fixed but the target policy changes from always-a_L to always-a_R; (B) the always-a_L question is fixed but its unobserved outcome changes; or (C) both remain fixed at value 0 while estimation error changes the report to 1. The latent causes differ, but the observable record is identical. □

An identity-blind rule therefore receives the same input in cases that require different conclusions. Resolving question identity is necessary, but it is not sufficient for causal attribution. We record identity, audit configuration, and append-only lifecycle state separately:

$$\begin{aligned} C_{\mathrm{id}}(q) &= (\text{id}, \text{version}, q, \text{parent}), \\ C_{\mathrm{audit}} &= (P_{\mathrm{ref}}, B, \mu, \phi, D_s, \epsilon, \text{probe/resampling version}), \\ R_t(q) &= (\text{status}, \text{evidence}, \text{probe outcomes}, \text{decision}, t). \end{aligned} \tag{4}$$

Together, these compact records distinguish an asserted question revision from an answer revision. Diagnosing world and estimation changes still requires controlled interventions, uncertainty estimates, or trusted diagnostics.

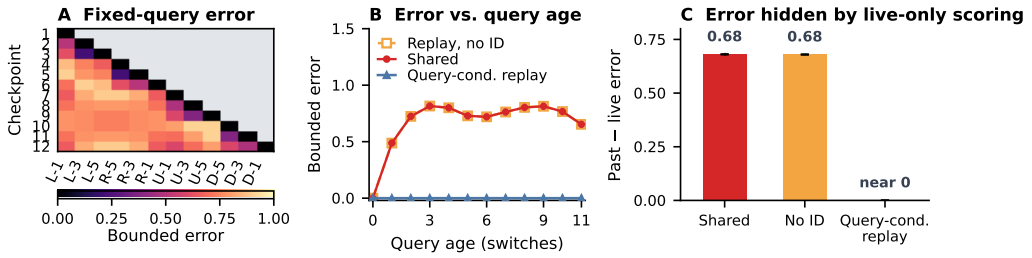


Figure 2: Pre-specified 12-query stream (10 held-out seeds). (A) Shared-head fixed-question error (gray: unseen). (B) Historical error versus query age. Conditioned replay is the completeness control. (C) Past minus live error over checkpoints 2–12. Metric: $\frac{1}{2}$(terminal-state + collision 0–1 error). Bars show seed standard errors.

4 Controlled Stress Tests

We use a deterministic grid world to evaluate every frozen historical query exactly. Twelve pre-specified queries span four actions and horizons $\{1, 3, 5\}$, with horizon-only and intervention-only switches. We score live and frozen queries after each 600-update phase. Methods share a base architecture and loss but differ in replay and query-conditioning inputs (Appendix A.2).

Across checkpoints 2–12 and 10 held-out seeds, bounded live/historical error is $0.0000/0.681$ for the shared head and $0.0005/0.680$ for replay without identity. Query-conditioned replay reaches near-zero bounded error ($0.0000/ < 0.001$), so the finite panel is jointly learnable (Figure 2). For both identity-blind learners, historical error exceeds live error at all 11 checkpoints. Live-only scoring thus hides frozen-question failure. This is not merely catastrophic forgetting: scoring only the live query replaces a fixed historical target with a moving one, making failure appear as successful adaptation. Appendix A.3 gives a controlled P/q/estimation-change existence check.

5 Audit Protocol: Evaluation Before Comparison

Attach an immutable specification and version to every answer. A change creates a successor ID, whose scoped compatibility is tested on reference anchors. Compare the same ID directly. Compare distinct IDs only through registered audit projections or supported warm starts, and abstain when unresolved. Identity is exact versioning; comparability is scoped. Record P_{ref}, the realized probe set B, sampling rule μ, ϕ, D_s, ϵ, and anchors. Under a memory budget, retain identity certificates, audit configurations, and lineage even after answers are discarded. Index an answer by question version, reference anchor, and declared readout, not only by checkpoint or output head.

Once comparability is established, factorial tests can vary P or q, or vary the learner while holding both fixed. We report four evaluation dimensions:

Fixed-question error: loss on frozen question versions.
Identity collision: distinct specifications assigned one ID.
Missed compatibility: verified compatible successors unnecessarily cold-started.
Attribution: correct P/q/learner labels under controlled arms.

Combined and video-scale arms remain proposals. Continual-RL settings can freeze specifications or grounded rollouts and precommit interventions [3, 11, 26]. Appendix A.5 gives a reporting card.

6 Limitations and Outlook

These tests establish possibility, not prevalence, in deployed video world models. Preserving stable high-dimensional reference anchors across tokenizer, encoder, and interface updates remains open. A central systems question is whether continual representation updates can preserve such anchors without freezing adaptation. If a specification or anchor is lost, the protocol abstains. Future benchmarks should report question versions, audit scopes, and historical-question error.

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A Minimal demonstrations and reproducibility

A.1 Analytic GVF special case

In the history-based formulation of Schlegel et al. [17], a GVF question $q_{\mathrm{GVF}} = (\pi, c, \gamma)$ defines the random continued return

$$G_t = \sum_{k \geq 0} \left(\prod_{j=1}^{k} \gamma(H_{t+j}) \right) c(H_{t+k+1}), \qquad V(P, q_{\mathrm{GVF}}; h) = \mathbb{E}_{P,\pi}[G_t \mid H_t = h].$$

In this instance, $Z_q = G_t$, $u = \pi$, and τ carries the continuation and discount semantics induced by γ. Hence $\mathfrak{L}(P, q_{\mathrm{GVF}}; h) = \mathcal{L}_{P,\pi}(G_t \mid H_t = h)$ and F is expectation. Identity remains auditable only if policy π, cumulant c, continuation γ, and their referenced implementations are immutable or versioned. The Alberta Plan and classical GVF formulations motivate this treatment of predictive questions as explicit objects. Here, ι extends it to learned interfaces whose token, channel, or action semantics can change across checkpoints. Strong semantic equivalence would require equal projected laws over a declared environment class $\mathcal{P}$; finite observations establish only scoped compatibility. The population counterpart of Eq. (2) is $d_{\mu,\phi}^{P_{\mathrm{ref}}}(q, q') = \mathbb{E}_{h \sim \mu} D_s(\phi_q(\mathfrak{L}(P_{\mathrm{ref}}, q; h)), \phi_{q'}(\mathfrak{L}(P_{\mathrm{ref}}, q'; h)))$. The analytic check uses one recurrent state and two immutable one-step GVF questions, q_L and q_R. Their Bernoulli target signals have means 0.2 and 0.8, and continuation terminates after the first target. Each phase lasts 400 online updates with constant step size 0.05. A shared scalar and identity-blind replay receive no question input, whereas the indexed control maintains one scalar per immutable question version. Table 1 reports absolute error after the second phase over 200 fixed seeds. Identity-blind replay approaches a compromise because its buffer combines incompatible targets without identifying the questions that generated them.

Table 1: Analytic special case (mean absolute error $\pm$ standard error; 200 seeds).

Method	Frozen old q_L	Live current q_R
Shared head	0.593 ± 0.005	0.053 ± 0.003
Replay, no identity	0.292 ± 0.005	0.308 ± 0.005
Question-indexed estimates	0.052 ± 0.003	0.053 ± 0.003

A.2 Learned action-conditioned rollout model

Two-query sanity check. The environment is an 11×11 grid with boundary walls and one internal wall opening. From a uniformly sampled free state, $q_{L,3}$ executes left for three steps and $q_{R,5}$ executes right for five. The predictor has two 128-unit ReLU layers with separate terminal-state and collision heads. For the conditioned method, inputs concatenate a one-hot state, a padded one-hot action sequence, and horizon. Unconditioned baselines receive zeros for the latter two fields. Each phase uses 600 Adam updates with learning rate 2×10^{-3} and batch size 96. During phase two, replay methods replace 20% of each batch with old-query examples. The loss is

$$\mathcal{L} = \mathrm{CE}(\widehat{s}_{t+\tau}, s_{t+\tau}) + \mathrm{BCE}(\widehat{z}_{\mathrm{collision}}, z_{\mathrm{collision}}),$$

where CE and BCE denote categorical and binary cross-entropy, respectively, and each term has unit weight. Evaluation draws $4|\mathcal{S}_{\mathrm{free}}|$ states with replacement using fixed seed 12345. Table 2 reports the values shown in Figure 3.

Table 2: Learned rollout loss (mean $\pm$ standard error; 10 seeds).

Method	Frozen old $q_{L,3}$	Live current $q_{R,5}$
Shared head	24.250 ± 0.421	0.0014 ± 0.0001
Replay, no identity	2.490 ± 0.012	0.335 ± 0.005
Query-conditioned replay	0.0018 ± 0.0002	0.0038 ± 0.0006

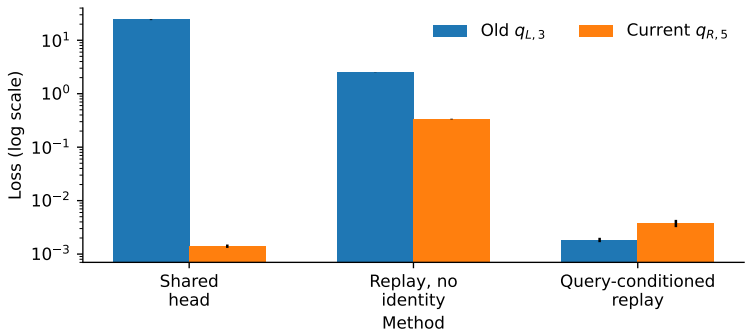


Figure 3: Frozen-old/live-current rollout loss (10 seeds; mean $\pm$ standard error; log scale). Replay uses 20% old samples; only conditioned replay receives the action script and horizon.

The shared head's frozen loss is large because phase-two optimization can drive the predicted probability of the old terminal state arbitrarily close to zero. Cross-entropy is unbounded, so 24.25 is a valid loss, not a normalized error rate or a numerical cap violation.

Twelve-query continual panel. The continual panel uses $q_{a,h}$ for $a \in \{\text{left}, \text{right}, \text{up}, \text{down}\}$ and $h \in \{1, 3, 5\}$, in the fixed order L1, L3, L5, R5, R3, R1, U1, U3, U5, D5, D3, D1. All phases use the architecture, optimizer, batch size, and 600-update budget above. Replay replaces 20% of each batch with examples from one uniformly sampled earlier question. The two replay methods use the same precomputed old-question schedule. Every checkpoint evaluates all 12 questions on the deterministic free-state panel, although unseen questions are excluded from retention metrics. For checkpoint $t > 1$, live loss is $\ell_t(q_t)$ and past loss is $\frac{1}{t-1}\sum_{i<t} \ell_t(q_i)$. Query order, hyperparameters, and acceptance thresholds were fixed before execution. Seeds 0–1 were used only for smoke testing; reported seeds 10–19 are disjoint. Alongside cross-entropy, we report bounded error $E = \frac{1}{2}(\mathbf{1}[\hat{s} \neq s] + \mathbf{1}[\hat{z} \neq z]) \in [0, 1]$, which averages terminal-state and collision classification errors. The predeclared gates in the supplementary README require finite values and each directional result at at least 8 of 11 checkpoints. Both cross-entropy and E pass at all 11 checkpoints: identity-blind past error exceeds live error, and conditioned replay reduces past error. Mean bounded live/past errors are $0.000000/0.680500$ (shared), $0.000500/0.680100$ (replay without identity), and $0.000000/0.000020$ (conditioned replay). The displayed zeros are finite-panel classification errors; the corresponding cross-entropy losses remain nonzero.

A.3 Magnitude-matched three-arm audit

The latent rollout model has 64-dimensional state and layout encoders, a gated recurrent-unit action-dynamics module, and terminal-state and collision decoders. It trains for 2,600 Adam updates on random one-to-five-step action sequences across eight wall layouts, with learning rate 2×10^{-3}, batch size 128, and loss $\mathrm{CE} + 2\,\mathrm{BCE}$. On a fixed panel of 66 states, the baseline query executes three rightward actions. The world arm adds a wall while holding the query and learner fixed. The question arm holds the world and learner fixed but replaces the baseline with a stochastic target policy that mixes three- and four-step rightward rollouts. The learner arm holds the world, query, encoder, and dynamics fixed while changing only a scalar collision-decoder calibration.

Five pilot seeds set the policy mixture (0.820) and calibration coefficient (0.519) to match the pilot world-arm report change. Both values were then frozen for 30 disjoint evaluation seeds. True and reported answer changes are absolute changes in mean collision probability over the fixed 66-state panel. True answer changes are 0.227 for the world intervention, 0.186 for the stochastic question intervention, and exactly 0 for learner calibration, while learned reports are intentionally magnitude-matched. The output-only observer is a deliberately narrow leave-one-seed-out nearest-centroid classifier over scalar change. Its 95% interval is cluster-bootstrapped over the 30 seeds. We do not test multi-anchor or time-series diagnosis. The audit checks a wall probe, immutable policy certificate, and parameter checksum. Its perfect controlled score follows by construction: it shows the sufficiency of registered side information, not automatic recovery.

Table 3: Held-out three-arm stress test (30 seeds). Arm changes are mean ± standard error; attribution uses a seed-cluster 95% confidence interval.

Measurement	Result
World-only reported change	0.205 ± 0.004
Question-only reported change	0.204 ± 0.002
Learner-only reported change	0.208 ± 0.002
True answer change (world/question/learner arm)	0.227/0.186/0.000
Output-only source attribution	35.6% (95% CI: 25.6–45.6%; chance 33.3%)
Registered side-information check	90/90 resolved by construction

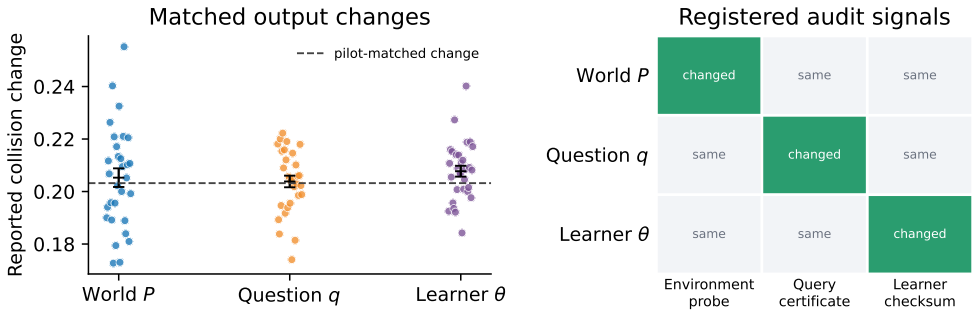


Figure 4: Magnitude-matched reports under isolated world, question, and learner interventions. Scalar-only attribution is near chance; registered side information identifies the constructed source (30 held-out seeds).

A.4 Identity collision and missed compatibility

Distinct left and right questions initially have equal Bernoulli means of 0.5. They are probe-compatible in the initial reference environment but retain separate IDs. A value-only identity merge creates an *identity collision*: after a world change moves their means to 0.2 and 0.8, any constructed merged estimate between them has analytic average MAE 0.3. This shows that compatibility need not transfer beyond P_{ref}. Conversely, a successor specification uses a separately versioned implementation while preserving the declared audit projection and mean 0.2 in P_{ref}. Cold-starting it is a *missed compatibility*. Reusing its estimate gives 0.052 rather than 0.110 MAE over the first 50

updates, and the ordering is preserved at 25 and 100 steps. These checks use one reference state, an identity audit functional, symmetric absolute mean difference, step size 0.05, 400 updates per phase, and 200 seeds. They show that scoped agreement can support reuse without establishing identity.

Table 4: Identity and compatibility checks (MAE $\pm$ standard error; 200 seeds).

Case	Method	MAE
Identity collision	Constructed merged estimate	0.300 (analytic)
Identity collision	Separate question IDs	0.050 ± 0.002
Missed compatibility	Syntax-only cold start	0.110 ± 0.002
Verified compatibility	Reused estimate	0.052 ± 0.002

A.5 Question Continuity Card

Table 5 gives a compact audit record for a cross-checkpoint prediction panel. It separates immutable identity from compatibility evidence and mutable lifecycle state.

Table 5: Question Continuity Card.

Field group	Required record
Identity certificate	Immutable question identifier/version, specification, parent, creation checkpoint
Query	Conditioning/history selector κ, intervention u, target object and readout $Y = (Z, F)$, horizon/termination τ, grounding/interface version ι
Audit configuration	Reference environment or suite, realized probe set B, sampling distribution μ, functional ϕ, discrepancy D_s, tolerance ϵ, probe version, resampling rule/unit, rollout count
Anchors	Environment-native actions and decodable observations; sensor, tokenizer, decoder, and actuator versions
Lifecycle record	Append-only active/superseded/retired status, evidence, probe outcomes, audit decision (compatible/incompatible/unresolved), timestamp
Budget	Retained answer or compact summary, storage/evaluation cost, anticipated control value

At a new checkpoint, an unchanged immutable specification preserves identity; any specification change creates a successor ID. When compatibility matters, run the declared audit projections in P_{ref}. An upper bound at most ϵ supports comparison or warm-starting only within the registered scope, a lower bound above ϵ rejects compatibility, and overlap remains unresolved. Probe results never merge IDs, and compatibility does not transfer to a new environment without a registered cross-environment test. If the declared reference anchor is unavailable, the protocol abstains whenever compatibility must be established.

The anonymous supplementary archive contains the exact scripts and generated result files for every demonstration.